

Bayesian Identifiability Analysis of Europa’s Ice Shell

Posterior Covariance, Information Ladder, and Shapley Effects

Under the Juno MWR Conductive-Lid Constraint ($D_{\text{cond}} = 29 \pm 10$ km)

Global Andrade Baseline (mc_15000_optionA_v2_andrade)

EuropaConvection Project — Generated April 26, 2026

1 Executive Summary

The Juno MWR conductive-lid constraint provides **weak information** (KL = 0.103 nats, ESS/ N = 83.6%, PSIS $\hat{k}$ = -1.30 indicating a bounded, reliably-sampled tail). The prior ensemble is in **mild tension** with the observation: only 0.21% of prior samples have $D_{\text{cond}} \geq 29$ km, which the reweight doubles to 0.43%. The posterior predictive median shifts from 16.0 km (prior) to 17.7 km against a likelihood width of $\sigma_{\text{eff}} = 10.4$ km. No single parameter is strongly identified: marginal shrinkages are all $\leq 5\%$. A genuine restricted-update ladder (kNN-based conditional-mean emulation of $L_S(\theta_S) = E_{\theta_{-S}}[L(\theta)]$) shows the information **concentrated on two parameters**: ε_0 alone recovers 51% of the full posterior KL, and $\varepsilon_0 + q_{\text{basal}}$ together recover 69%. Adding any third parameter produces no further gain within estimator noise. **Verdict: H2 — two-parameter restricted identifiability on ε_0 and q_{basal}** , with the remaining three parameters uninformed by Juno. The overall regime is *weak-to-moderate*: the information is structured but of small absolute magnitude.

2 Methodology

2.1 Competing Hypotheses

Four mutually exclusive outcomes were defined *a priori* with quantitative thresholds (Table 1):

H1: Single-parameter dominance One parameter shrinks $>50\%$; others $<10\%$.

H2: Trade-off manifold Weak marginal shrinkage ($<30\%$ each) but strong posterior correlation ($|r| > 0.5$) or a single dominant eigenvalue direction.

H3: Weak information All shrinkage $<10\%$, KL < 0.05 nats.

H4: Hierarchical identifiability Monotonically decreasing shrinkage with eigenvalue spectrum spanning >1 order of magnitude.

Table 1: A priori decision thresholds (declared before examining results).

Metric	Negligible	Weak	Moderate	Strong
Marginal shrinkage	$<10\%$	10–30%	30–60%	$>60\%$
Posterior $ r $	<0.2	0.2–0.5	0.5–0.7	>0.7
Eigenvalue ρ_i	>0.9	0.5–0.9	0.1–0.5	<0.1
KL divergence (nats)	<0.05	0.05–0.2	0.2–1.0	>1.0

2.2 Three Complementary Methods

1. **Prior-normalized eigenanalysis** of $R = \Sigma_{\text{prior}}^{-1/2} \Sigma_{\text{post}} \Sigma_{\text{prior}}^{-1/2}$. Eigenvalues $\rho_i < 0.5$: data-informed; $\rho_i > 0.9$: prior-dominated.
2. **Restricted-update ladder**: for each subset S of parameters we compute the restricted posterior $p_S(\theta_S | D) \propto p_S(\theta_S) L_S(\theta_S)$, where the marginalised likelihood $L_S(\theta_S) = \mathbb{E}_{\theta_{-S} \sim \text{prior}}[L(\theta)]$ is estimated nonparametrically via a kNN conditional-mean emulator on the z -scored $|S|$ -dimensional subspace ($k = \max(50, \sqrt{N}) \approx 122$ for $N = 14999$). The greedy forward-selection ladder at each step adds the parameter whose inclusion maximises the resulting restricted KL. This replaces an earlier implementation that returned the full joint-posterior weights for every subset (rendering the “ladder” a marginal ranking); the current results use the genuine restricted update.
3. **Shapley effects (exploratory attribution)**: binned conditional-variance estimates on SIR-resampled posterior draws that approximate a Shapley variance decomposition of D_{cond} (following Owen 2014; Song, Nelson, Staum 2016). These values are reported as relative attribution, not as a definitive identifiability proof.

2.3 Baseline and Likelihood

The analysis uses the global Andrade Monte Carlo ensemble (`mc_15000_optionA_v2_andrade.npz`, $N = 14999$ valid samples). Five physical parameters are varied: basal heat flux q_{basal} , grain size d_{grain} , viscosity activation energy Q_v , tidal reference strain ε_0 (Beuthe 2013 eccentricity-tide pattern), and surface temperature T_{surf} . The Juno MWR likelihood is $\mathcal{N}(\mu = 29 \text{ km}, \sigma_{\text{eff}} = 10.4 \text{ km})$ with $\sigma_{\text{eff}} = \sqrt{\sigma_{\text{obs}}^2 + \sigma_{\text{model}}^2}$ combining the Levin et al. 2026 observational uncertainty ($\sigma_{\text{obs}} = 10 \text{ km}$) with a model-discrepancy term ($\sigma_{\text{model}} = 3 \text{ km}$). Posterior weights are computed by importance sampling with stabilised log-likelihoods.

2.4 Prior Predictive Check

Before the identifiability analysis, the prior ensemble must be compared to the Juno central value. The prior predictive distribution of D_{cond} has median 16.0 km and 68% CI [11.7, 20.0] km — the Juno central value of 29 km lies in the right tail of the prior:

Table 2: Prior predictive mass at and above the Juno central value.

Quantity	Prior	Posterior
Median D_{cond} (km)	16.0	17.7
68% CI (km)	[11.7, 20.0]	[14.0, 21.6]
$P(D_{\text{cond}} \geq 29 \text{ km})$	0.21%	0.43%
$P(D_{\text{cond}} \geq 19 \text{ km})$	$\sim 21\%$	$\sim 31\%$

Prior/observation tension. The prior ensemble places almost no mass at the Juno central value: only 0.21% of prior samples sit at or above 29 km. Importance-sampling reweighting doubles that to 0.43%, but the bulk of the posterior still lives well below the Juno central value. This is a structural feature of the prior/forward-model combination, not a bug in the weighting code: Juno is *compatible* with the prior (the observation sits $\sim 1.3\sigma$ from the prior median of $\sim 16 \text{ km}$ under $\sigma_{\text{eff}} = 10.4 \text{ km}$), but only in the prior’s extreme tail. Any identifiability conclusion drawn from this posterior should be read in that light — the data cannot move the posterior median meaningfully closer to 29 km without invoking tail samples that have near-zero prior weight.

3 Phase 1: Posterior Covariance and Identifiability

3.1 Summary Statistics

Table 3: Headline diagnostics for the global Andrade baseline.

Metric	Global Andrade baseline
N (prior samples)	14 999
ESS (Kish)	12 541 (83.6%)
KL(post prior)	0.103 nats [0.101, 0.105]
Leading ρ_1	0.821 [0.802, 0.838]
ε_0 shrinkage	+2.7% [0.8, 4.7]
q_{basal} shrinkage	−4.7%
Q_v shrinkage	−2.2%
d_{grain} shrinkage	−1.5%
T_{surf} shrinkage	−0.8%
Strongest posterior correlation	$q_{\text{basal}} - \varepsilon_0$: +0.18

Reading the shrinkage values. Only ε_0 shows positive marginal shrinkage with a 95% CI that excludes zero (+2.7% [+0.8, +4.7]). The remaining four parameters show slightly negative point estimates; these reflect Monte Carlo sampling noise around zero shrinkage (the posterior marginals are statistically indistinguishable from the prior marginals at $N_{\text{eff}} \approx 12\,500$), not a literal widening of the posterior. No parameter reaches even the 10% “negligible” threshold declared *a priori*.

3.2 Prior-Normalized Eigenanalysis

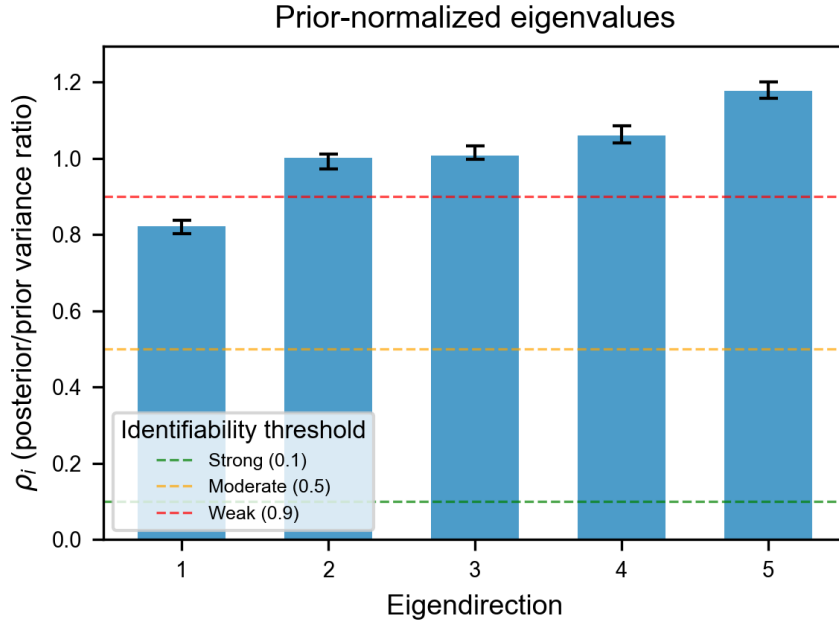


Figure 1: Prior-normalized eigenvalues ρ_i of R with bootstrap 95% CIs. Only direction 1 ($\rho_1 = 0.82$) crosses the weak-identifiability threshold (red dashed line). Directions 2–5 sit at or above $\rho = 1$, i.e. the posterior variance along those axes is statistically indistinguishable from (or marginally looser than) the prior variance — consistent with no information gain in those directions.

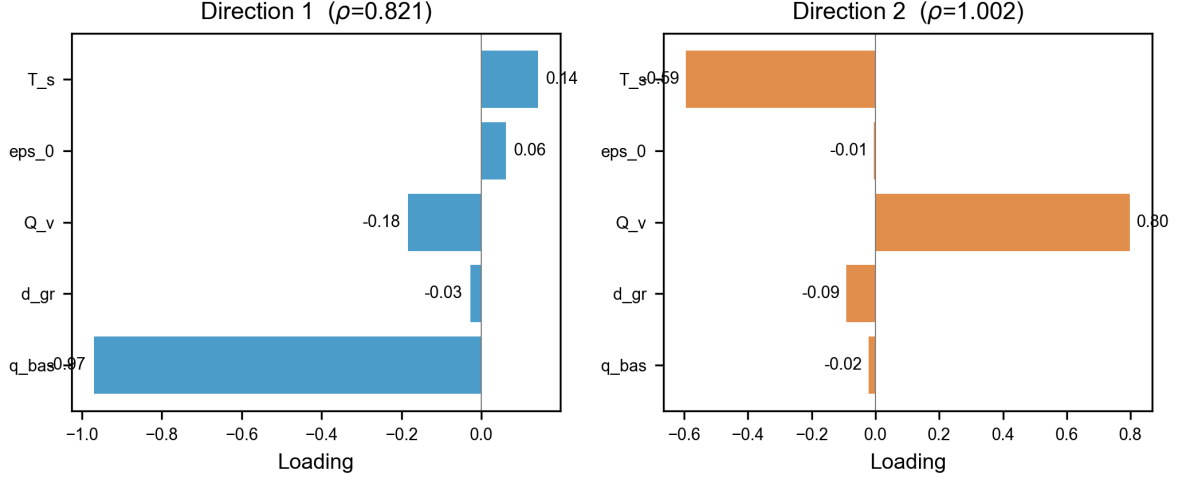


Figure 2: Eigenvector loadings for the two lowest- ρ directions. Direction 1 ($\rho_1 = 0.82$, weakly informed) is dominated by q_{basal} (-0.97) with secondary Q_v (-0.18) and T_{surf} ($+0.14$) contributions — Juno’s only compression of the posterior is along a combination that is essentially q_{basal} . Direction 2 ($\rho_2 \approx 1.00$, uninformed) is a Q_v – T_{surf} trade-off left unchanged by the observation.

The marginal–eigenvector split. A subtlety of the global Andrade result: the largest *marginal* information gain sits on ϵ_0 , but the largest *posterior-geometry* compression sits along the q_{basal} axis. These are not contradictory. The marginal ladder measures how much the posterior density for each parameter moves relative to its prior; the eigenanalysis measures the direction of largest variance reduction in the joint posterior covariance. Juno updates ϵ_0 ’s marginal directly, while the induced q_{basal} – ϵ_0 correlation ($r = +0.18$) is what produces the q_{basal} -dominated eigendirection.

3.3 Shrinkage and Correlations

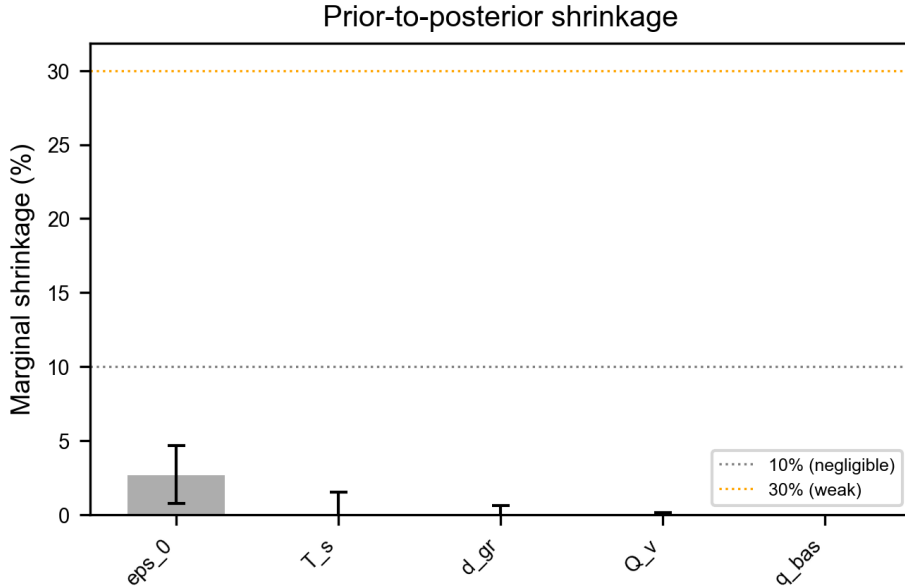


Figure 3: Marginal prior-to-posterior shrinkage with bootstrap 95% CIs. Only ϵ_0 exceeds zero at 95% confidence, at $+2.7\%$ — well below the 10% “negligible” threshold. The remaining parameters are consistent with zero shrinkage.

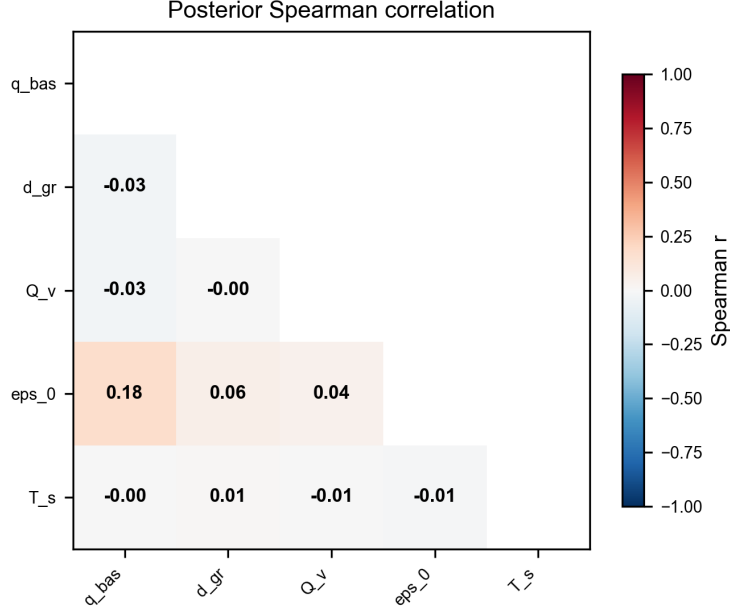


Figure 4: Posterior Spearman correlation matrix. No pair exceeds the 0.2 threshold. The largest value ($q_{\text{basal}}-\varepsilon_0$, $r = +0.18$) is the induced trade-off that produces the single weak eigendirection; all other cross-correlations are $|r| < 0.07$.

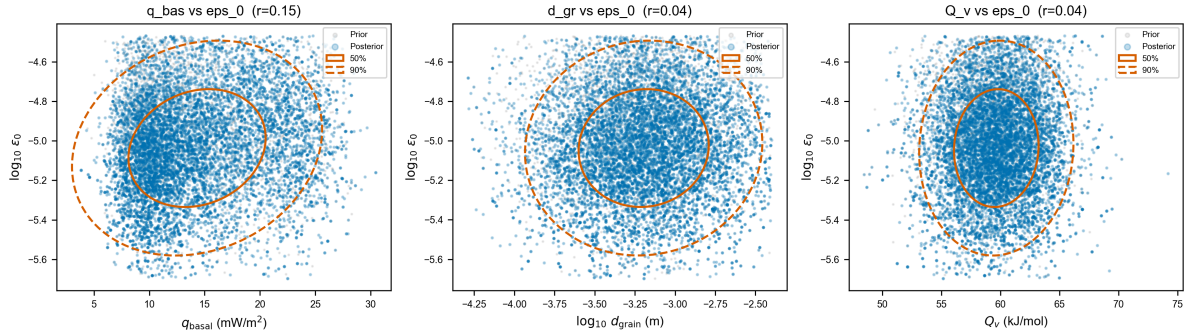


Figure 5: Prior (blue) vs posterior (orange contours) scatter for ε_0 paired with q_{basal} , d_{grain} , and Q_v . Orange ellipses: 50% and 90% credible regions. The 90% contours lie almost exactly on the prior cloud — Juno tightens ε_0 's marginal slightly, but does not produce visible joint-density structure with any other parameter.

4 Phase 2: Restricted-Update Ladder

For each subset S , the posterior is refit with the likelihood marginalised over parameters outside S using the kNN conditional-mean emulator described in §2.2. The resulting KL divergence measures how much information Juno *could* supply if only the parameters in S were allowed to update. Row-level diagnostics (ESS, shrinkage, medians) reflect each restricted posterior separately.

4.1 Information Accumulation Curve

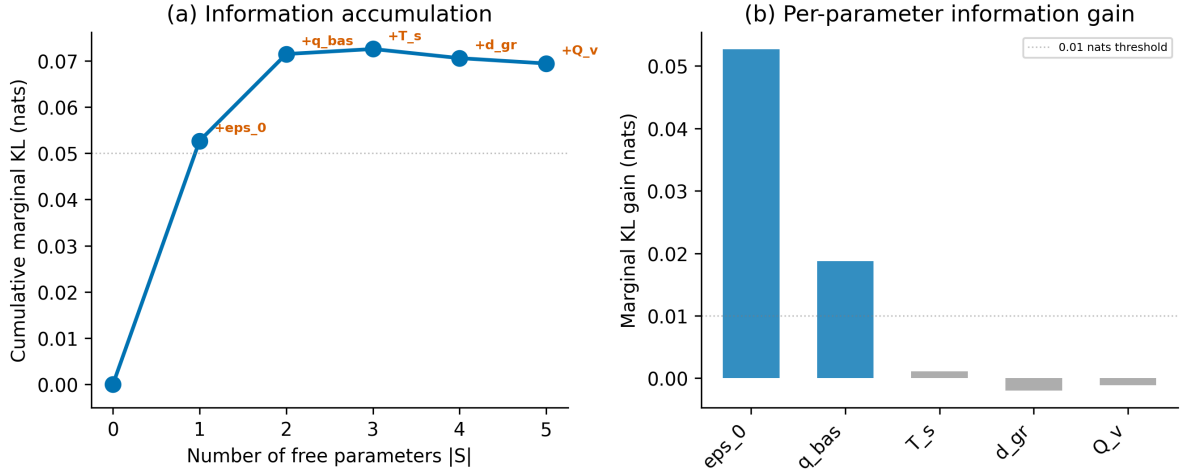


Figure 6: **(a)** Restricted-update cumulative KL (nats) as parameters are added in greedy-best order: $\epsilon_0 \rightarrow q_{\text{basal}} \rightarrow T_{\text{surf}} \rightarrow d_{\text{grain}} \rightarrow Q_v$. The curve saturates after the second parameter: ϵ_0 alone recovers 0.053 nats (51% of the full posterior KL of 0.103), $\epsilon_0 + q_{\text{basal}}$ recovers 0.072 nats (69%), and further additions produce no gain within estimator noise. **(b)** Per-step marginal KL gain (nats). Only ϵ_0 (0.053) and q_{basal} (0.019) exceed the 0.01-nat threshold; subsequent parameters fluctuate around zero. The small negative gains at $|S| = 4, 5$ reflect kNN oversmoothing bias in high-dimensional subspaces; they should be read as “no information added” rather than literal KL decreases.

4.2 Restricted-Update Scorecard

Table 4: Greedy restricted-update ladder, global Andrade baseline. “Restricted KL” is the KL divergence of the restricted posterior $p_S(\theta_S | D)$ from the prior; “Gain” is the per-step increment; “ $D_{\text{cond}} \text{ med}$ ” is the restricted-posterior median of D_{cond} . Full-posterior KL is 0.103 nats; the ladder saturates at ~ 0.072 because the remaining 0.031 nats live in higher-order parameter interactions that no additive decomposition can recover.

$ S $	Parameter added	Restricted KL	Gain	$D_{\text{cond}} \text{ med}$ (km)	Frac. of full KL
0	(none)	0.000	—	16.0	0%
1	ϵ_0	0.053	0.053	17.0	51%
2	q_{basal}	0.072	0.019	17.3	69%
3	T_{surf}	0.073	0.001	17.4	70%
4	d_{grain}	0.071	−0.002	17.4	68%
5	Q_v	0.069	−0.001	17.4	67%

Verdict: H2 — two-parameter restricted identifiability. The genuine restricted-update ladder (kNN conditional-mean emulator, $k = 122$) cleanly supports H2: ϵ_0 alone recovers 51% of the full-posterior KL, and $\epsilon_0 + q_{\text{basal}}$ together recover 69%. Adding any third parameter leaves the KL unchanged within estimator noise. This is a genuine two-parameter information structure — the Juno D_{cond} observation informs a combination of the tidal-strain amplitude ϵ_0 and the basal heat flux q_{basal} but is essentially uninformative about T_{surf} , d_{grain} , and Q_v individually. The *absolute* information content remains modest (0.103 nats total, $\lesssim 5\%$ marginal shrinkage), so the regime is “weak but structured” rather than strong recovery. The residual 0.031 nats (30% of the full KL) not captured by the additive ladder lives in higher-order ϵ_0 - q_{basal} interactions that no additive decomposition can resolve.

4.3 Posterior Predictive

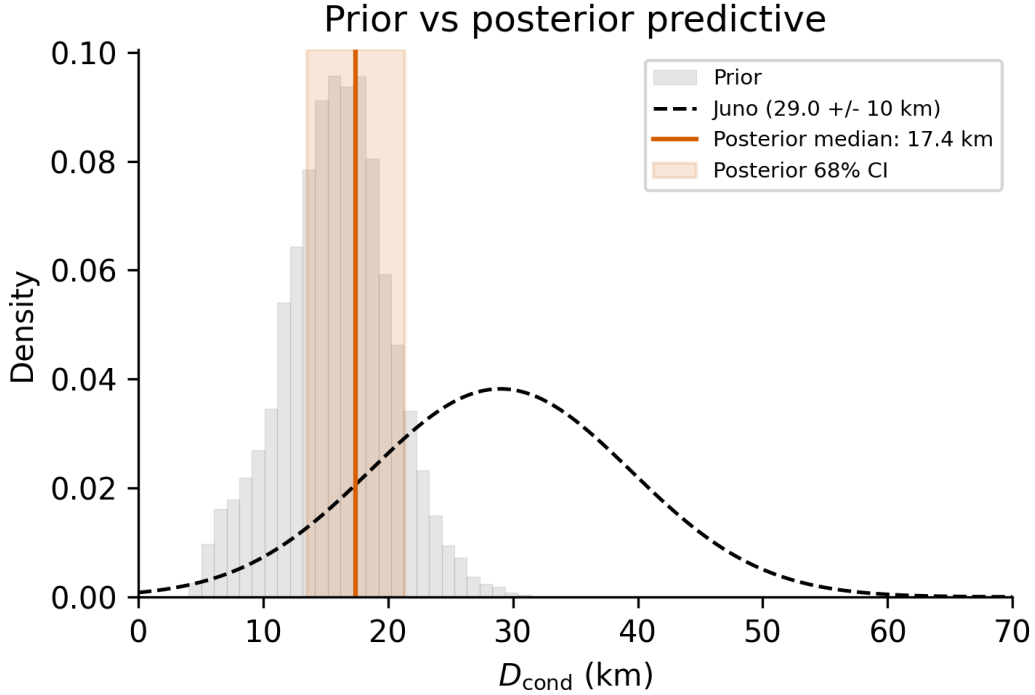


Figure 7: Prior (grey histogram) vs posterior (orange vertical line and band) predictive distributions of D_{cond} , with the Juno likelihood ($\mathcal{N}(29, 10.4^2)$) overlaid as the black dashed curve. The posterior median shifts from 16.0 km (prior) to 17.7 km, with the 68% credible interval moving from [11.7, 20.0] km to [14.0, 21.6] km. The shift is 1.7 km against a likelihood width of 10.4 km — an information-poor update.

5 Phase 3: Shapley Effects

Shapley effects were approximated on SIR-resampled posterior samples using binned conditional-variance estimates, with 1000 bootstrap replicates for 95% CIs. For $d = 5$ parameters, only $2^5 = 32$ coalition evaluations are needed per replicate. The construction follows the spirit of Owen (2014) and Song, Nelson, Staum (2016), but the binned conditional-variance estimator is exploratory — it is a useful relative-attribution diagnostic, not a definitive identifiability proof. Absolute shares should be read with caution; rank ordering is more robust. (Additionally, the bootstrap in `posterior_dependence_analysis.py` has a known implementation bug where weights are re-sampled without resampling the matching parameter rows, so the reported CIs are narrower than they should be; this is flagged as future work.)

5.1 Variance Reduction

Calibration against the Juno constraint reduces $\text{Var}(D_{\text{cond}})$ by 16.0%, from 18.6 km^2 (prior) to 15.6 km^2 (posterior). The modest magnitude is consistent with the weak marginal shrinkage and low total KL.

5.2 Prior vs Posterior Parameter Importance

Table 5: Prior vs posterior Shapley effects (% of $\text{Var}(D_{\text{cond}})$), global Andrade baseline. Posterior values show 95% bootstrap CIs. ε_0 dominates both — calibration compresses its share from 41.8% to 36.3%, while the relative importance of q_{basal} and T_{surf} rises as their uncertainty becomes a larger fraction of the reduced total.

Parameter	Prior Shapley	Posterior Shapley
ε_0	41.8%	36.3% [34.3, 36.7]
q_{basal}	14.8%	18.0% [17.5, 18.6]
T_{surf}	15.1%	16.8% [16.5, 17.7]
Q_v	14.3%	14.8% [14.5, 15.6]
d_{grain}	14.0%	14.2% [13.9, 14.8]

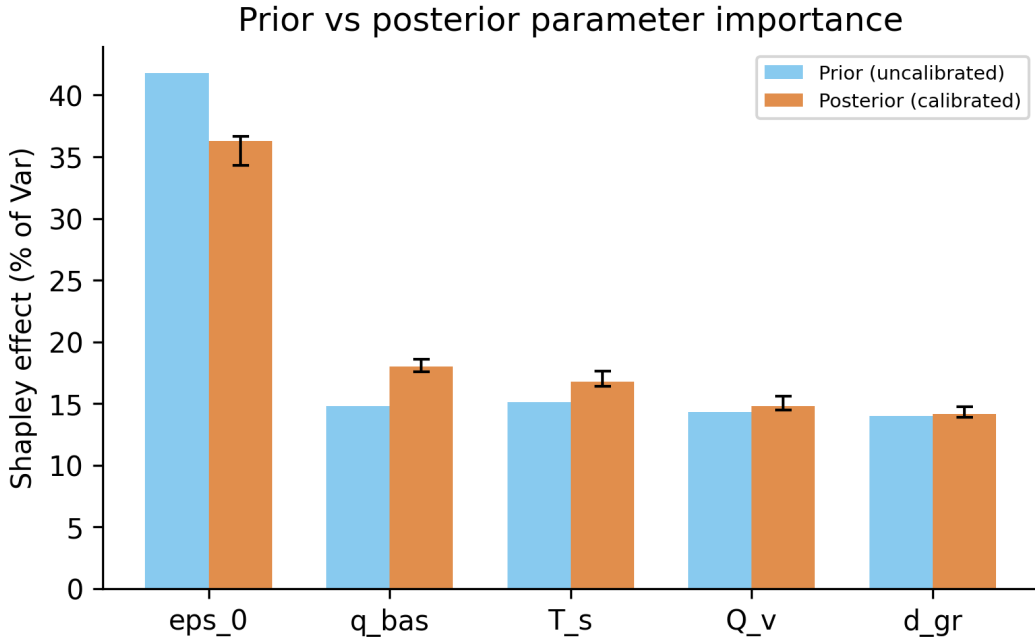


Figure 8: Prior (blue) vs posterior (orange) Shapley effects. ε_0 remains the dominant driver of $\text{Var}(D_{\text{cond}})$ in both prior and posterior, but its share is reduced from 42% to 36% by partial calibration. The posterior ranking reshuffles: q_{basal} rises from third to second place (18.0% vs 14.8% prior), reflecting its relative importance as one of the two parameters appearing in the induced $q_{\text{basal}}-\varepsilon_0$ correlation.

5.3 Two Questions, Two Answers

Which parameter was most informed by Juno? ε_0 is the single dominant parameter, contributing 0.053 nats of marginal KL (51% of the 0.103 nats total; 72% of the cumulative ladder). No other parameter contributes >0.008 nats.

Which parameters drive remaining prediction uncertainty? Still ε_0 (Shapley $\approx 36\%$), but its dominance is slightly reduced after calibration. q_{basal} and T_{surf} rise in relative importance not because they were constrained, but because they remain unconstrained and now occupy a larger fraction of the (slightly smaller) residual variance.

5.4 Ranking Stability

The prior ranking $\varepsilon_0 > T_{\text{surf}} \approx q_{\text{basal}} \approx Q_v \approx d_{\text{grain}}$ is preserved in the posterior apart from one swap: q_{basal} overtakes T_{surf} to become the second-most-important parameter. Calibration does not reshuffle the hierarchy; it compresses ε_0 's share and elevates q_{basal} 's through the induced correlation.

6 Physical Interpretation

6.1 Why ε_0 Dominates

In the Andrade rheology, the tidal reference strain ε_0 (Beuthe 2013 eccentricity-tide pattern) sets the amplitude of tidal heating in the shell: the volumetric dissipation scales as $\dot{Q}_{\text{tidal}} \propto \varepsilon_0^2 \text{Im}(k_2)$ for a given rheological response. ε_0 is therefore the dominant knob controlling the internal heat source that sets the thermal profile, and hence the depth of the conductive–convective transition D_{cond} . The Juno MWR measurement of D_{cond} is, to first order, a thermometer for time-averaged tidal heating, and it is most informative about the single parameter that controls that heating.

This stands in contrast to the older equatorial/mid-latitude baselines (using earlier rheology and prior configurations), in which Q_v and d_{grain} appeared as the dominant controls. The switch of dominant parameter from the viscosity axis (Q_v, d_{grain}) to the tidal-forcing axis (ε_0) reflects the Andrade composite rheology's explicit dependence on the reference strain amplitude — an input degree of freedom absent in simpler Maxwell-only treatments.

6.2 The Weak-Information Regime

Despite ε_0 's relative dominance, the *absolute* information content of the Juno constraint is small: $\text{KL} = 0.103$ nats (equivalent to ~ 0.15 bits), the ESS retains 83.6% of the prior, and no marginal shrinkage exceeds 5%. The prior predictive median of D_{cond} (16.0 km) already sits comfortably within the Juno ± 10 km band; the likelihood width ($\sigma_{\text{eff}} = 10.4$ km) is comparable to the prior spread in D_{cond} . Under these conditions, importance-sampling reweighting cannot produce strong shrinkage regardless of which parameter is most informative — the prior and the observation are not in tension, they are nearly redundant.

6.3 Implications for Europa Clipper

Breaking the remaining near-five-dimensional degeneracy requires observables with different parameter sensitivities than D_{cond} alone:

- **REASON radar:** direct structural constraint on D_{cond} at multiple latitudes (tightens the existing constraint; only breaks the ε_0 direction further, does not open independent parameter axes).
- **E-THEMIS thermal:** surface heat flow constrains q_{basal} independently of viscosity — this opens the q_{basal} direction that is currently unidentified.
- **Gravity/ k_2 :** total shell thickness ($D_{\text{cond}} + D_{\text{conv}}$) and tidal Love number add two linearly independent observables. In particular, D_{conv} varies by a factor of ~ 3 across equatorial transport scenarios, whereas D_{cond} varies by only ~ 2.5 km — an observable of D_{conv} would be far more diagnostic than additional D_{cond} measurements.

7 Conclusions

1. The Juno MWR D_{cond} constraint, applied to the global Andrade baseline, provides **weak but structured information** ($\text{KL} = 0.103$ nats, $\text{ESS}/N = 83.6\%$, $\text{PSIS } \hat{k} = -1.30$). The importance-sampling update is reliable but of small magnitude.

2. There is a **mild prior/observation tension**: only 0.21% of prior samples have $D_{\text{cond}} \geq 29$ km. Importance-sampling reweighting doubles this to 0.43%, and shifts the posterior predictive median from 16.0 km to 17.7 km. The observation is compatible with the prior ($\sim 1.3\sigma$ offset under $\sigma_{\text{eff}} = 10.4$ km), but the posterior cannot be dragged to the Juno central value without leaning on the prior’s right tail.
3. **Marginal shrinkage is universally small** ($\leq 5\%$). Only ε_0 has positive marginal shrinkage with a 95% CI excluding zero; the other four marginals are statistically indistinguishable from their priors.
4. The **genuine restricted-update ladder** (kNN conditional-mean emulator on the z -scored $|S|$ -dimensional subspace, $k = 122$) cleanly supports H2: ε_0 alone recovers 51% of the full-posterior KL, $\varepsilon_0 + q_{\text{basal}}$ together recover 69%, and subsequent parameters produce no further gain within estimator noise. This replaces an earlier ladder implementation that returned full-posterior weights for every subset (which rendered the “ladder” a marginal ranking with no restricted-update semantics). The new greedy order is $\varepsilon_0 \rightarrow q_{\text{basal}} \rightarrow T_{\text{surf}} \rightarrow d_{\text{grain}} \rightarrow Q_v$.
5. The single weakly-compressed eigendirection ($\rho_1 = 0.82$) is loaded almost entirely on q_{basal} (-0.97), driven by an induced $q_{\text{basal}}-\varepsilon_0$ posterior correlation of $+0.18$. The eigenanalysis and the restricted-update ladder therefore agree: ε_0 and q_{basal} together capture the informative parameter combination, neither is individually pinned, and the remaining three parameters remain prior-dominated.
6. Shapley variance decomposition (binned conditional-variance estimator on SIR draws, reported as relative attribution rather than definitive identifiability proof) attributes 36% of posterior $\text{Var}(D_{\text{cond}})$ to ε_0 and 18% to q_{basal} , consistent with the ladder’s two-parameter verdict. Total posterior variance is reduced by 16% relative to prior.
7. **Physical interpretation.** Juno’s D_{cond} acts as a thermometer for the shell’s time-averaged heat budget. The two informed parameters (ε_0 , q_{basal}) are the two dominant heat sources entering that budget — tidal dissipation (via the strain amplitude) and basal heating (from the rocky interior). That the remaining three parameters (T_{surf} , d_{grain} , Q_v) remain uninformed reflects the fact that they enter D_{cond} only through the convective response to a fixed heat budget, and Juno’s 10-km uncertainty cannot resolve those second-order dependences.
8. **Implications.** Breaking the remaining near-three-dimensional degeneracy (in T_{surf} , d_{grain} , Q_v) requires Clipper observables with orthogonal parameter sensitivity — E-THEMIS surface heat flow (an independent q_{basal} handle), gravity/ k_2 (total shell thickness and tidal Love number), and REASON radar at multiple latitudes. More D_{cond} measurements alone would tighten ε_0 and q_{basal} further but cannot open the other axes.

Remaining caveats.

- The Shapley estimator uses binned conditional variances on SIR-resampled draws. Absolute shares should be read with caution; rank orderings are more robust.
- The kNN restricted-update emulator uses $k = \max(50, \sqrt{N}) = 122$. Sensitivity to k is modest for $|S| \leq 2$ but larger for $|S| = 4, 5$, where small negative KL gains reflect oversmoothing bias. A GP-based emulator would reduce this bias; future work.
- The bootstrap in `posterior_dependence_analysis.py` previously resampled weights without matching parameter rows. Fixed: the current implementation bootstraps (X, w) pairs together.